

SIT225 Data Capture Technologies

Assessment Pass Task 3

Critical Reflection & Evidence-Informed Revision

Estimated time: 10-12 hours

Last updated: 14 June 2026

Due: End of Week 7

Hurdle: This task contributes to the Learning Portfolio milestone for Week 7. Submission required to stay on track for unit completion.

I. Instructions and Conditions

This task is due in **Week 7 (Sunday)**, and you are expected to start early. If your first submission is incomplete or incorrect, you may resubmit after receiving feedback. There will be **no extensions**, and late submissions will not be accepted. If your submission is not correct and complete by the deadline, it will be marked as a **FAIL**. This task is a **hurdle requirement**, failure to submit a correct version on time will result in failing the unit.

All submissions will be checked for **plagiarism** and **GenAI** use. *Any GenAI usage should be acknowledged (see template below) and you must include, in appendix, the prompt, raw response, and your changelog how you have adopted the GenAI response in your work.* You must complete the work **independently**, and you must not share or show your work to anyone else, unless it is explicitly instructed in your task to do so. If you have questions or encounter problems, attend the on-campus or online classes, or use the **Discussion Board** on the unit site.

Per Deakin policy, include an Acknowledgements section:

"This assessment work was developed with assistance from the following approved Deakin genAI tools:

- Deakin GEM (accessed June 2026): Used to <generate structured critique on protocol design and confounding variables (3 iterations)>.
- Microsoft Copilot for web (accessed June 2026): Used to <summarise ASHRAE comfort standards and identify relevant literature.>

All genAI-generated suggestions were critically evaluated and integrated only where supported by evidence. The final design, decisions, and justifications are my own."

II.Task Overview

This task builds on *assessment pass task 2* (research-informed use case design & ethical analysis) by introducing a *peer review and AI critique cycle*. You will:

1. *Solicit structured peer feedback* on your Assessment 2 design
2. *Obtain AI-assisted critique* using approved Deakin tools
3. *Produce a critical response report* evaluating all feedback
4. *Justify acceptance/rejection decisions* with evidence from literature and theoretical frameworks

This task assesses:

- ULO2: Apply professional and ethical standards
- ULO3: Design protocols meeting user and functional requirements
- ULO1 (implicit): Critically analyse and interpret feedback

Key alignment: Develops *evaluative judgement* through peer dialogue and reflective practice, consistent with Deakin's curriculum framework.

III.Task Components

Part A: Peer Feedback Solicitation

Engage peers to critique your Assessment 2 design before you submit this task.

A.1 Peer Review Protocol

You will be matched with your peer by week 7 so you can review each other's Assessment 2 submission (use case, sensors, protocol, hypotheses, ethics). You should engage early enough and share your assessment-2 proposal PDF to receive response in time.

A sample structured review template is shown below:

<i>Review Criterion</i>	<i>Peer Feedback Prompt</i>	<i>Response space</i>
Use case clarity	Is the problem statement specific and researchable? Could you replicate the study design from the description?	[peer writes 100–150 words]
Literature review rigour	Did the student cite credible sources ($\geq 2-3$)? Is the research gap clearly distinguished from the original problem?	[feedback]
Hypothesis quality	Are hypotheses testable and grounded in literature? Could you predict how you'd measure each one?	[feedback]
Sensor selection	Are sensors justified for the problem? Any gaps or redundancies?	[feedback]
Ethical considerations	Did the student address privacy, consent, and data minimisation? Are there overlooked stakeholders?	[feedback]

<i>Review Criterion</i>	<i>Peer Feedback Prompt</i>	<i>Response space</i>
Feasibility	Is the protocol realistic for a student to execute in 10+ weeks? Any technical or logistical concerns?	[feedback]
Strongest element	What is the most compelling aspect of the design?	[feedback]
One improvement	What single change would strengthen the design most?	[feedback]

Part B: GenAI-Assisted Critique (Using Approved Tools)

Obtain structured AI feedback using Deakin-approved genAI tools.

B.1 Approved Tools for This Task

Per Deakin policy, use *one or more approved tools*:

<i>Tool</i>	<i>Best Use</i>	<i>How to Access</i>
Deakin GEM	Clarify concepts; ask for critique on design logic; reflect on strengths/gaps	Deakin GEM landing page
Studiosity+	Get feedback on written clarity of your Assessment 2 design document	Via Deakin website or CloudDeakin
Microsoft Copilot for web	Brainstorm alternative approaches; summarise relevant literature	Microsoft 365 login

Important guardrails:

- Do NOT enter personal/confidential data
- GenAI may generate incomplete or incorrect responses - critically evaluate all outputs
- Use genAI to *support your thinking*, not replace it
- Acknowledge your use in your response report (see Part C)

B.2 Sample GenAI Prompts (for Deakin GEM or Copilot)

Use these prompts (or adapt) to solicit structured critique:

Prompt 1: Protocol Design Critique

"I've designed a sensor protocol to test if seasonal temperature variation and occupancy interact to affect comfort ratings. I'm using DHT22 (temp/humidity) + PIR (motion) sensors, collecting data for 3 months. My hypotheses are: [paste your 2–3 hypotheses]. What are potential weaknesses in this design? What confounding variables might I have missed? Suggest 2–3 improvements."

Prompt 2: Ethical Gap Identification

"I'm collecting temperature, humidity, and occupancy data from [your location: classroom/room/building]. My privacy plan includes: [paste your privacy measures]."

What ethical or privacy issues might I have overlooked? Who else might be affected by this data collection? Suggest one additional safeguard."

Prompt 3: Literature Validation

"I found these research gaps in my literature review: [paste your gap statement]. My hypotheses aim to address this gap. Are there existing methods or theoretical frameworks I should consider? Suggest 2–3 related papers or concepts I should explore."

Prompt 4: Statistical/Methodological Soundness

"I plan to test my hypothesis using [your method: e.g., linear regression, t-test, classification]. My sample size will be [duration/readings]. What statistical assumptions should I verify? Will my sample size be sufficient?"

B.3 GenAI Feedback Compilation

Document AI critique in a *GenAI feedback summary*:

<i>Prompt / Topic</i>	<i>GenAI Response (key points)</i>	<i>Tool used</i>	<i>Date accessed</i>
Protocol design critique	"Potential confound: HVAC response lag varies by building. Consider logging actual HVAC on/off times."	Deakin GEM	June 2026
Ethical gaps	"Missing: informed consent language for occupants. Data retention policy vague (delete when?)."	Deakin GEM	June 2026
Literature validation	"Consider Fanger's PMV model and recent work on occupant behaviour (ASHRAE). Your gap aligns with 'real-time comfort prediction' literature."	Copilot	June 2026

Part C: Critical Response Report (1200–1500 words)

Write a *structured response* that critically engages with *both peer and AI feedback*.

C.1 Introduction (150–200 words)

- Briefly state your Assessment 2 use case and key design choices
- Overview: How many peers reviewed (only 1 here)? Which genAI tools did you use?
- Purpose: To iteratively strengthen design through evidence-informed critique

Example:

"My Assessment 2 design proposes a 3-month study of how seasonal temperature variation and occupancy patterns interact to affect occupant comfort in a university classroom. I collected feedback from a single peer and used Deakin GEM and Copilot to identify design gaps and ethical considerations. This report critically evaluates 12 feedback points and justifies my response."

C.2 Feedback Summary & Thematic Analysis (300–400 words)

Synthesise peer and AI feedback by *theme*, not individual reviewer:

2.1 Themes from Peer & AI Feedback

Identify 3–4 major themes across all feedback:

Theme	Peer feedback	AI feedback	Consensus?
Confounding variables	"Consider circadian rhythm & lighting effects"	"HVAC lag & building thermal mass unaccounted for"	YES—multiple sources identify same concern
Data retention ethics	"Privacy notice vague on deletion policy"	"When will you delete raw data?"	YES—both flag specificity gap
Sample size adequacy	"3 months may not capture inter-seasonal variation"	"Consider 12-month minimum for robust seasonal patterns"	CONFLICT—peer thinks sufficient; AI suggests longer
Hypothesis clarity	"H2 unclear operationalisation"	"H2 about occupancy prediction; measurement method not specified"	YES—hypothesis needs refinement

Write 300–400 words synthesising these themes:

"Across 3 peer reviews and 4 genAI critiques, three major themes emerged: (1) **Confounding variables:** Peers noted lighting and circadian effects; genAI highlighted HVAC response lag. Combined, these suggest my 'temperature → comfort' pathway is oversimplified. I initially assumed HVAC response is immediate, but AI critique correctly notes lag could be 10–30 min, confounding occupancy-temperature relationships. (2) **Ethical specificity:** All sources flagged my vague privacy notice. Peers said 'when deleted?' and genAI asked 'raw vs. aggregated data retention?' This gap is significant. (3) **Temporal adequacy:** Peer A thought 3 months sufficient; genAI suggested 12 months for robust seasonal patterns. Literature review (ASHRAE comfort standards) supports year-round data for seasonal comparison."

2.2 Identify Conflicts or Tensions

Explicitly note where feedback disagrees:

"**Conflict example:** Peer feedback suggested PIR occupancy detection sufficient (binary on/off); genAI critique noted nuance—occupant dwell time matters (1 min vs. 30 min), not just presence. This is valid; my protocol will need duration thresholds."

C.3 Critical Evaluation: Accept / Reject / Revise (600–800 words)

For each major feedback point, state your decision and justify with evidence.

Format:

<i>Feedback Point</i>	<i>Your decision (Accept / Reject / Revise)</i>	<i>Justification with evidence</i>	<i>Literature / source</i>
<i>Confounding: HVAC lag</i>	Revise	Original protocol assumes instant HVAC response. AI critique valid. HVAC systems typically have 5–20 min lag (thermal mass). Accepting this means I must log HVAC on/off events separately. I revise H1 to account for lag: "Comfort correlates with temperature trajectory, adjusted for HVAC lag phase."	ASHRAE 90.1 standards [cite]; genAI Deakin GEM (June 2026) Acknowledging y... Students
<i>Data retention: Delete timeline</i>	Accept	Peer & AI both flagged vague retention. Accepting feedback: I commit to deleting raw sensor readings after 30 days; retain aggregated daily averages for 90 days; discard all data post-analysis (6 months). Privacy notice updated (Appendix A).	Privacy-by-Design principle (ISO 27001) [cite source]; Deakin Assessment 2 ethics framework
<i>Sample adequacy: 3 vs. 12 months</i>	Reject (with caveat)	12-month data is ideal but infeasible for Week 9 submission. However, I accept AI critique partially: I will aim for 4–5 months (Jan–May) capturing early-to-mid year variation. This is compromise between rigour and feasibility. I will note in limitations: "Seasonal conclusions limited; winter (July–Aug) data absent."	Feasibility constraints vs. rigour trade-off; transparent limitation reporting aligns with ULO2 ethics
<i>Hypothesis H2 operationalisation</i>	Revise	Peer & AI both noted H2 unclear ("occupancy predicts waste"). I revise: "H2: Rooms vacant >15 min drift >1°C from setpoint; HVAC predictive activation (15-min pre-occupancy trigger) reduces drift to <0.5°C." Now testable, measurable.	Refined from peer & genAI input [acknowledgement in Appendix B]

Write justified narratives for each point (600–800 words total). (*format example not shown here*)

C.4 Protocol Revision Summary (200–300 words)

Concisely state *how your Assessment 2 design will change* based on accepted feedback:

Original design:

- 3 months data; DHT22 + PIR; H1: temperature → comfort
- Privacy notice: "Data retained for analysis"

Revised design (post-feedback):

- 4–5 months data (Jan–May); DHT22 + PIR + light sensor; H1: temperature + HVAC lag → comfort (with confound controls)
- Privacy notice: "Raw data deleted 30 days post-collection; aggregates retained 90 days; final deletion 6 months post-analysis"
- Hypothesis H2 operationalised: Occupancy-predictive HVAC trigger reduces thermal drift

C.5 Reflection: Evaluative Judgement & Professional Growth (200–300 words)

Reflect on the feedback process itself. How did peer and AI critique improve your thinking?

Prompts:

- Which feedback surprised you or challenged an assumption?
- How does accepting/rejecting feedback reflect your professional values (ethics, rigour, feasibility)?
- What would you do differently next time?

Example reflection:

"Initially, I dismissed the 12-month suggestion as infeasible. However, genAI's prompt - 'seasonal conclusions are weak with 3 months'—made me reconsider. I realised I was optimising for deadline rather than rigour. Peer feedback grounded this: one peer noted the study was 'publishable if methods are transparent about limitations.' This reframed my approach: accept shorter timeline, but be ruthlessly honest about limitations. This aligns with ULO2 (ethical standards) [1]—transparent communication of constraints, not pretence of certainty. Going forward, I'll distinguish 'ideal' from 'feasible' study designs earlier."

C.6 Acknowledgement of GenAI Use

Per Deakin policy, include an *Acknowledgements* section:

"This response report was developed with assistance from the following approved Deakin genAI tools:

- *Deakin GEM (accessed June 2026): Used to generate structured critique on protocol design and confounding variables (3 iterations).*
- *Microsoft Copilot for web (accessed June 2026): Used to summarise ASHRAE comfort standards and identify relevant literature.*

All genAI-generated suggestions were critically evaluated and integrated only where supported by evidence or peer consensus. The final design, decisions, and justifications are my own."

Part D: Supporting Documentation

Attach to your response report:

D.1 Peer Feedback Forms (unsigned is OK if privacy preferred)

- A single completed peer review templates
- Note any identifying info removed per privacy

D.2 GenAI Interaction Records

- Screenshots or export of genAI prompts + responses
- Annotate where you agreed/disagreed with AI suggestions
- OR: Paste key excerpts into Appendix A

D.3 Revised Assessment 2 Design (Appendix B)

- 1-page summary of changes (side-by-side: original vs. revised)
- Updated privacy notice
- Revised hypotheses

IV.Submission Checklist

<i>Component</i>	<i>Format</i>	<i>Length</i>	<i>Notes</i>
Week 6-7 activities	Word or PDF	--	Perform each week's activity and export PDF to finally combine together with other items mentioned below. Clearly set heading of each week and activity number.
<i>Part A: Peer feedback</i>	Compiled table + original forms	--	1 peer review; signed/dated
<i>Part B: GenAI critique</i>	Summary table + interaction records	--	Screenshots or text excerpts; tool/date logged
<i>Part C: Response report</i>	PDF (main document)	1200–1500 words	Includes intro, synthesis, decisions (accept/reject/revise), revision summary, reflection, acknowledgement
<i>Part D: Supporting docs</i>	Appendices (A & B)	--	GenAI transcripts; revised protocol summary
<i>Consolidate each part of the task into a single PDF, append previous weeks' (week 6-7) activity sheets at the bottom of the PDF. There are no code or data to share for this assessment.</i>			

Total package: ~1500 words + appendices; ≤10 pages

V.Grading Criteria (Pass/Fail)

<i>Criterion</i>	<i>Pass</i>	<i>Notes</i>
<i>Feedback quality</i>	1 substantive peer review obtained; genAI	Authentic engagement; not

<i>Criterion</i>	<i>Pass</i>	<i>Notes</i>
	tools used appropriately	fabricated reviews
<i>Critical analysis</i>	Feedback synthesised into 3–4 clear themes; conflicts identified; nuance shown	Avoid "accept all" or "reject all" responses; show discernment
<i>Decision justification</i>	Each accept/reject/revise decision supported by evidence (literature, feasibility, ethics, ULOs)	Citations present; logic transparent; not arbitrary
<i>Protocol revision</i>	Original Assessment 2 design clearly differs from revised version; changes directly address feedback	Appendix B shows before/after
<i>Reflection & growth</i>	Reflection section demonstrates learning; connects to ULOs & professional practice	Honest, thoughtful, not generic
<i>genAI acknowledgement</i>	Tool names, dates, and use case clearly stated	Complies with Deakin policy
<i>Writing quality</i>	Clear structure, APA citations, Australian English, professional tone	Accessible; well-organised

Pass awarded when: All criteria met. Emphasis on critical engagement and justified reasoning.

VI.Support Resources

Approved genAI tools:

- Deakin GEM: <https://www.deakin.edu.au/students/study-support/study-resources/artificial-intelligence/approved-genai-tools-for-learning/deakin-gem>
- Studiosity+: <https://www.deakin.edu.au/students/study-support/study-resources/study-assistance/studiosity>
- Microsoft Copilot: <https://m365.cloud.microsoft/chat>

Guidance on acknowledging genAI:

- <https://www.deakin.edu.au/students/study-support/study-resources/artificial-intelligence/acknowledging-your-use>

Feedback integration guide:

- Deakin Study Support: "Responding to Feedback"

Literature & frameworks:

- ASHRAE 55 (thermal comfort standards)
- ISO 27001 (privacy-by-design)
- Assessment (Higher Education Courses) procedure

VII.Common Pitfalls to Avoid

- Fabricating peer feedback (verify with authentic peer signatures/contact)

- Over-relying on genAI without critical evaluation; treating it as "truth"
- Rejecting all feedback without justification ("my design is fine as-is")
- Accepting all feedback uncritically (show discernment; feasibility matters)
- Forgetting to acknowledge genAI use (required by Deakin policy)
- No connection to literature; decisions unsupported
- Response report reads as "defensive" rather than reflective